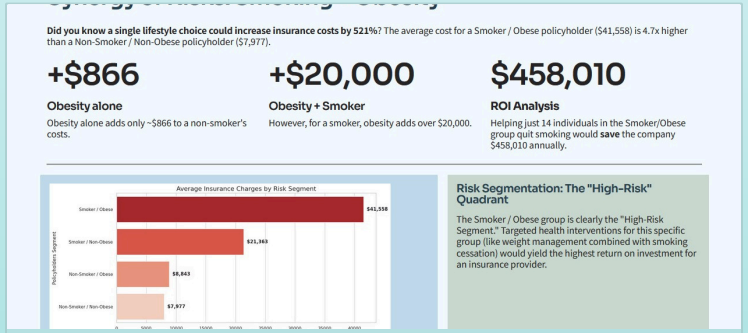
LARYSA GALAGAN

Optimizing Risk-Based Pricing for AN Health Insurance

Analyzing Medical Cost Drivers and Developing Predictive Models:

Leveraging data from 1,338 policyholders to transition from flat-rate pricing to a data-driven risk model.



GitHub

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📈 Insurance Cost Analysis & Segmentation Data-driven study of medical...

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Business Case & Objectives

AN Health Insurance (a fictional company) is a regional provider facing rising medical claim costs. In a competitive market, they cannot simply raise premiums across the board without losing customers to cheaper rivals. They need a data-driven approach to understand cost drivers and implement a "Risk-Based Pricing" model.

The Problem

The organization lacks a granular understanding of why certain policyholders incur 4x higher costs than the average. Without this insight, low-risk customers are overcharged (and leaving), while high-risk customers are underpriced (causing financial "leakage").

The Scale

Approximately 20% of the policyholder base accounts for nearly 52% of total charges.

Objectives

1. Identify the top 3 drivers of costs.
2. Build a predictive model to estimate individual charges.
3. Define a "High-Risk Segment" (standard threshold BMI > 30 + Smoker) and calculate the potential savings if 10% of this group joins a smoking cessation program.

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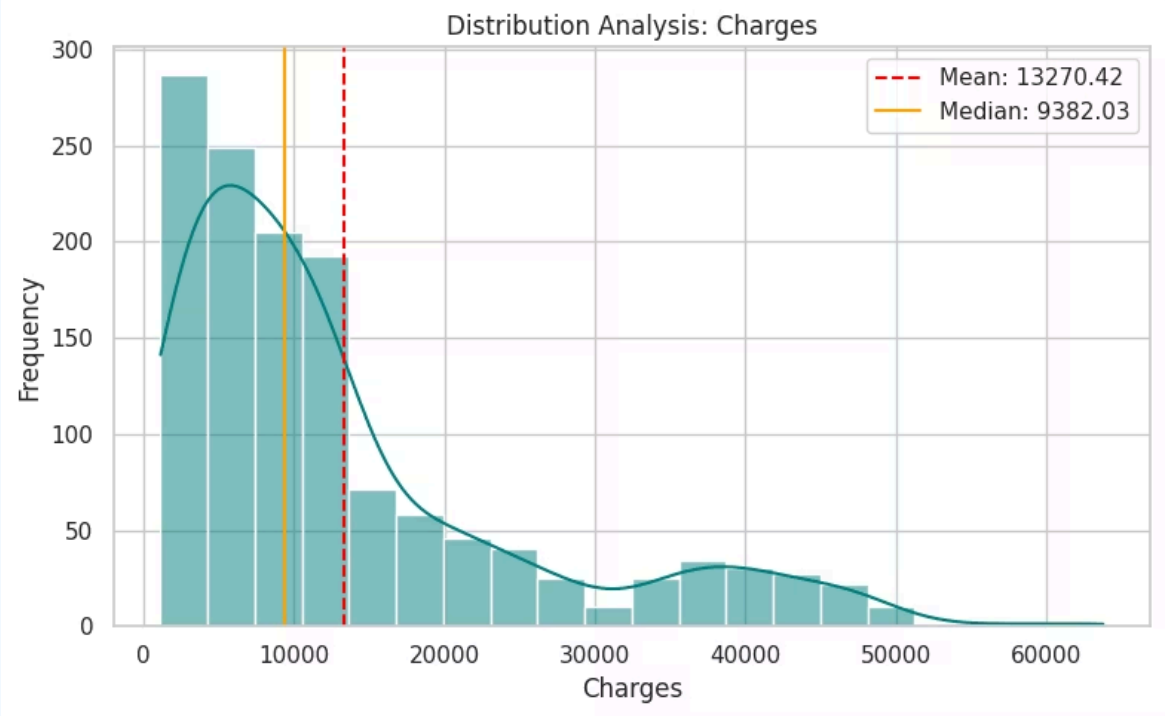
Who are our Policyholders?



Charges

Mean charge is **$13,270**, but the distribution is highly right-skewed, driven by high-cost outliers.

Demographics

Balanced distribution across gender and regions.

Age

Average age is **39**, with a notable spike in the young adult segment (18-20). Our dataset has an **over-representation of young adults**.

Geographical location of the policyholders

The company's clients are almost evenly distributed across all four regions, with only slightly more policyholders in the southwest.

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The Primary Driver: Smoking Status

Correlation

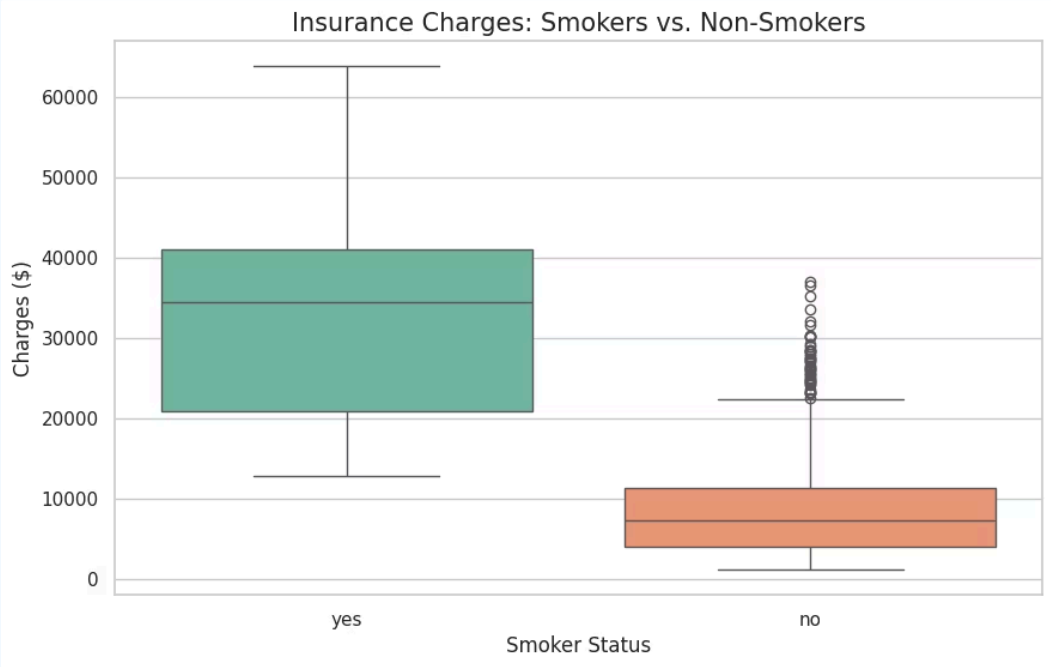
Smoking has a high positive correlation of **0.79**, indicating it is the strongest predictor of high charges in this dataset.

Cost Gap

On average, smokers are charged **$32,050**, while non-smokers are charged only **$8,434**.

Insight

A smoker's medical charges are nearly **4x higher** than those of a non-smoker. This suggests that smoking cessation programs could be the most effective way to reduce overall insurance payouts.



The data shows a massive disparity in costs based on smoking habits.

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The "Smoker" Multiplier

Three bands

We can clearly see the data splits into three distinct, upward-sloping bands:

The Bottom Band (Blue): This represents non-smokers. Their charges are the lowest and show the least variance.

The Middle Band (Mixed): This contains non-smokers with higher costs (likely due to other health factors like BMI or pre-existing conditions) and some smokers.

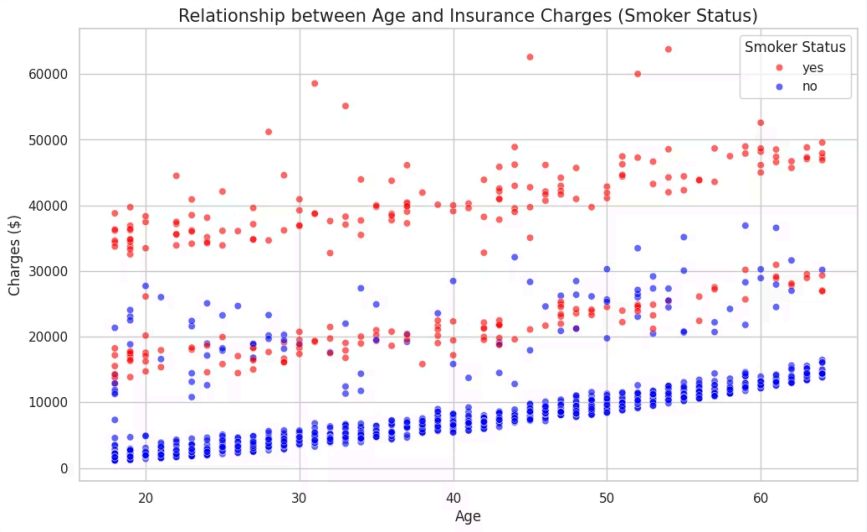
The Top Band (Red): This is exclusively smokers. Their charges are significantly higher, starting at a baseline that is often double or triple that of non-smokers in the same age bracket..

Non-Smokers

Exhibit a highly predictable, low-variance increase in costs as they age, generally staying below the $15,000 threshold until their late 50s.

Smokers

Experience a massive "step-up" in premiums. Even the youngest smokers (age 18–20) often face charges higher than the oldest non-smokers (age 60+).



Smoking acts as a vertical multiplier. While a 20-year-old non-smoker might pay under $5,000, a 20-year-old smoker is frequently charged between $15,000 and $40,000. The "gap" between the blue and red clusters represents the significant financial penalty (or risk premium) associated with smoking.

The data indicates that while age is a consistent driver of insurance costs, smoking status is the primary determinant of high-cost outliers.

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The Impact of Aging and Physical Metrics

Two other variables show a moderate **positive correlation with charges**:

- **Age** (0.30): As age increases, insurance charges tend to rise, which aligns with standard actuarial expectations. Seniors pay 83% more than the youngest group.
- **BMI** (0.20): There is a slight positive correlation between Body Mass Index and charges, suggesting that higher BMI contributes to higher costs, though less significantly than smoking or age..

Factor	Strength of Impact	Key Takeaway
Smoking	Very High	Increases charges by ~$23,600 on average.
Age	Moderate	Costs rise significantly between "Young" and "Senior" groups.
BMI	Low/Moderate	Influences costs, but less than lifestyle and age.
Region/Sex	Negligible	These factors are not meaningful predictors of cost.

Some variables show almost **zero correlation with insurance charges**: Sex, Number of Children, and Region.

However, there is a notable correlation of 0.27 between **BMI and the Southeast region**, indicating that individuals in this dataset from the Southeast tend to have a higher BMI than those in other regions.

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Synergy of Risks: Smoking + Obesity

Did you know a single lifestyle choice could increase insurance costs by 521%? The average cost for a Smoker / Obese policyholder ($41,558) is 4.7x higher than a Non-Smoker / Non-Obese policyholder ($7,977).

+$866

Obesity alone

Obesity alone adds only ~$866 to a non-smoker's costs.

+$20,000

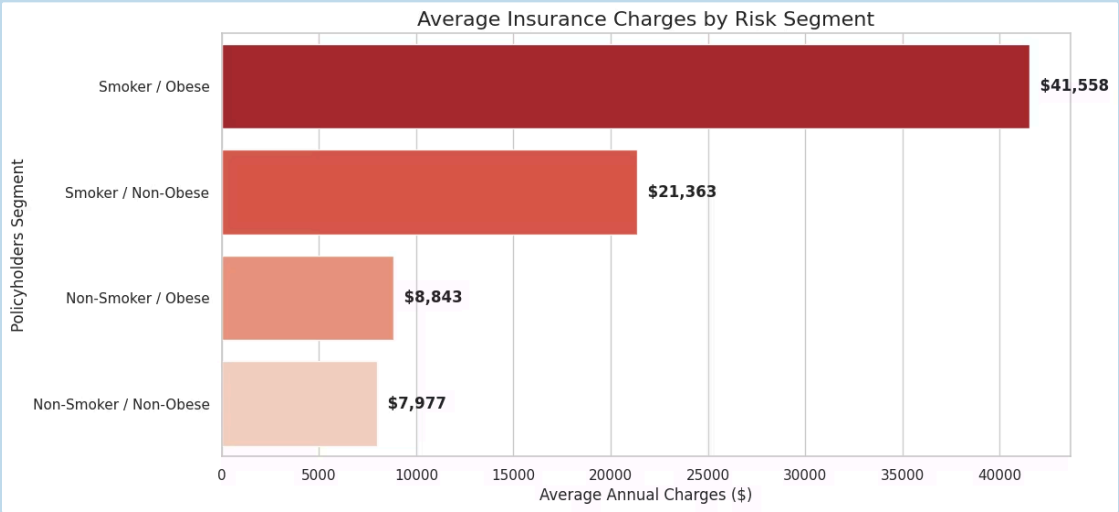
Obesity + Smoker

However, for a smoker, obesity adds over $20,000.

$458,010

ROI Analysis

Helping just 14 individuals in the Smoker/Obese group quit smoking would **save** the company $458,010 annually.



Risk Segmentation: The "High-Risk" Quadrant

The Smoker / Obese group is clearly the "High-Risk Segment." Targeted health interventions for this specific group (like weight management combined with smoking cessation) would yield the highest return on investment for an insurance provider.

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BMI-Smoker Interaction: Risk and Profitability

Smoking Status and Age are primary drivers of insurance costs

The data clearly suggests that lifestyle choices (smoking) and biological aging are the primary drivers of insurance costs. Health metrics like BMI play a secondary role, while demographic factors such as gender and geographic location are largely statistically insignificant in predicting total charges. To reduce insurance premiums according to this model, smoking cessation would be the most impactful variable.

However,

Predictive model

We have developed a predictive model that achieves **81.7% accuracy in estimating insurance charges**. The inclusion of a **BMI-Smoker interaction term** allowed the model to accurately **capture the high-risk synergy** between these two variables, which is the leading driver of outlier costs.

This model enables underwriters to **accurately price policies** and **protect profit margins** by capturing non-linear health risks

BMI and smoking status

The business should prioritize BMI and smoking status as the primary "**risk levers**" when calculating premiums.

Price policies

By reducing the variance between predicted and actual costs, the company can price policies more accurately, preventing "under-pricing" for high-risk individuals and improving overall profit margins.

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Data-Driven Action Plan

1 Pricing

Immediate 20% surcharge for smokers to cover their higher risk.

2 Wellness Program

- Target the "Southeast" region with obesity-prevention campaigns.
- A smoking cessation program for smokers.

3 Incentives

Introduce "Healthy BMI" discounts to attract and retain low-risk customers.

4 Ethics

Ensure pricing fairness for individuals with pre-existing medical conditions

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Technical Description

1,338 anonymized historical records.

Variable	Type	Description
age	int64	'female' 'male'
sex	object	age of policyholder
bmi	float64	Body Mass Index
children	int64	number of children
smoker	object	'yes' 'no'
region	object	'southwest' 'southeast' 'northwest' 'northeast'
charges	float64	annual insurance charges (USD)

- **Dataset:** https://www.kaggle.com/mirichoi0218/insurance
- **Language:** Python (Pandas, NumPy).
- **Modeling:** Scikit-Learn (Linear Regression).
- **Visualization:** Matplotlib/Seaborn for EDA.
- **Techniques:** Outlier Detection, One-Hot Encoding, R-squared Evaluation.

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